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Snake hibernacula

Abstract

A communal snake hibernaculum, including a communal chamber having a lower end below a ground level at a hibernaculum site with a water zone at the lower end to hold rehydration water and a snake hibernaculum life zone above the water zone. The communal chamber has a snake entrance providing snake access to the life zone and includes snake support shelves within the communal chamber at a plurality of vertically-spaced locations, the snake support shelves including at least one shelf in the life zone. A water outlet opens through a sidewall of the communal chamber above the lower end of the communal chamber and below the life zone to prevent flooding of the life zone, and the sidewall is a wall of a container installed at the hibernaculum site.

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Background/Summary

TECHNICAL FIELD
(1) The disclosure relates to snake hibernacula, and particularly to in-ground snake hibernacula.
BACKGROUND
(2) Snake overwintering survival in many areas requires a hibernaculum which provides the snake with a subterranean space that remains aerobic through the winter, does not freeze or flood completely, and is free of predators. Species and age differences result in different selections of overwintering positions, and many snakes are able to dive in water for at least short periods of time.
(3) Snake hibernacula have been built which provide subterranean spaces for overwintering. However, snake hibernacula of the prior art may be inefficient or ineffective. Accordingly, there is a need for an improved snake hibernacula.
SUMMARY
(4) According to some aspects, there is provided a communal snake hibernaculum, comprising a communal chamber installable at a hibernaculum site with a life zone within the chamber at a depth below a frost line and above a groundwater level, the communal chamber including a sidewall extending between a lower end and an opposing upper end with a water outlet opening through the sidewall between the lower end and the life zone to separate the life zone from a water zone at the lower end of the chamber and prevent flooding of the life zone when the communal chamber is installed at the hibernaculum site; a plurality of snake support shelves within the communal chamber at a plurality of positions spaced between the upper and lower ends of the chamber, including at least one snake support shelf within the life zone; and a snake entrance opening from an exterior of the communal snake hibernaculum into the communal chamber above the water outlet to provide snake access to the life zone.
(5) In some examples, the water outlet includes a passage extending between an intake end opening through the sidewall and a discharge end, the passage provided to carry water away from the hibernaculum when installed at the hibernaculum site.
(6) The communal snake hibernaculum may be installed at the hibernaculum site with the water outlet passage having an above-ground discharge end.
(7) The communal snake hibernaculum may further comprise a water inlet opening into the communal chamber above the lower end of the communal chamber to supply water to fill the water zone.
(8) The communal snake hibernaculum may be installed at the hibernaculum site with an intake end of the water inlet opening from a seasonally-available water supply to seasonally introduce new water to the communal chamber.
(9) The communal snake hibernaculum may be installed at the hibernaculum site with the bottom

end of the communal chamber extending below the ground level to a depth of at least 1 meter.
(10) The communal snake hibernaculum may be installed at the hibernaculum site with the upper end above the ground level and insulated by an overburden.
(11) At least a plurality of the snake support shelves may be formed by a plurality of blocks each having at least one internal cell opening through the block.
(12) The communal chamber may be sized to hold a plurality of snakes and has a volume of at least 7,500 cubic centimeters.
(13) The communal snake hibernaculum may be a multi-species hibernaculum and has a plurality of snake entrances including a first snake entrance sized for a first snake species and a second snake entrance sized for a second snake species, the first snake entrance being a different size than the second snake entrance.
(14) The communal snake hibernaculum may further comprise a monitoring well extending into the chamber from an exterior of the communal snake hibernaculum.
(15) According to some aspects, there is provided a communal snake hibernaculum, comprising a communal chamber having a lower end below a ground level at a hibernaculum site with a water zone at the lower end to hold rehydration water and a snake hibernaculum life zone above the water zone, the communal chamber having a snake entrance providing snake access to the life zone and including snake support shelves within the communal chamber at a plurality of vertically-spaced locations, the snake support shelves including at least one shelf in the life zone; and a water outlet opening through a sidewall of the communal chamber above the lower end of the communal chamber and below the life zone to prevent flooding of the life zone, and wherein the sidewall is a wall of a container installed at the hibernaculum site.
(16) In some examples, the container further comprises a bottom wall, and the sidewall and bottom wall form the lower end below the ground level.
(17) The container may be a plastic drum and the sidewall and bottom wall may be part of a unitary body.
(18) The life zone may be above a groundwater level at the hibernaculum site.
(19) The snake hibernaculum life zone may be at a depth that is below a frost line at the hibernaculum site.
(20) According to some aspects, there is provided a method of constructing a communal snake hibernaculum, comprising installing a container at a hibernaculum site to form a communal chamber with a subterranean water zone at a lower end of the chamber to hold rehydration water and an additional zone above the water zone and positioned to ordinarily remain aerobic, above a water level in the communal chamber, and above freezing such that the additional zone is a snake hibernaculum life zone; forming at least one snake entrance opening into the communal chamber from an exterior of the communal snake hibernaculum to provide snake access to the life zone; forming a water outlet opening through a sidewall of the container above the lower end and below the life zone to prevent flooding of the life zone; and forming a plurality of snake support shelves within the communal chamber at a plurality of vertically-spaced locations between the lower end and an upper end of the container, including at least one snake support shelf in the life zone.
(21) In some examples, installing the container at the hibernaculum site includes installing the container with an upper end of the container above the ground level, the method further comprising forming an overburden over the upper end of the container.
(22) A level of groundwater outside the communal chamber may fluctuate between a seasonal high level during a first season and a seasonal low level during a second season, and the water outlet opens into the communal chamber at a position that is above the seasonal low level.
(23) Forming the plurality of snake support shelves may include arranging cinder blocks within the communal chamber, each cinder block having at least one internal cell opening through the block.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The drawings included herewith are for illustrating various examples of systems, methods, and apparatus of the present specification. In the drawings:
(2) FIG. **1** is a side cross sectional view of a first communal snake hibernaculum, in accordance with an embodiment;
(3) FIG. **1**A is a top perspective view of a monitoring system of the hibernaculum of FIG. **1**;
(4) FIG. **2**A is side cross sectional view of a second communal snake hibernaculum, in accordance with an embodiment;
(5) FIG. **2**B is a top view of the hibernaculum of FIG. **2**A;
(6) FIG. **3**A is side cross sectional view of a third communal snake hibernaculum, in accordance with an embodiment;
(7) FIG. **3**B is a first top cross sectional view of the hibernaculum of FIG. **3**A;
(8) FIG. **3**C is a second top cross sectional view of the hibernaculum of FIG. **3**A;
(9) FIG. **4**A is side cross sectional view of a fourth communal snake hibernaculum, in accordance with an embodiment;
(10) FIG. **4**B is a top cross sectional view of the hibernaculum of FIG. **4**A;
(11) FIG. **5**A is side cross sectional view of a fifth communal snake hibernaculum, in accordance with an embodiment;
(12) FIG. **5**B is a top cross sectional view of the hibernaculum of FIG. **5**A;
(13) FIG. **6** is a top perspective cross sectional view of a sixth communal snake hibernaculum, in accordance with an embodiment;
(14) FIG. **7** is a top perspective cross sectional view of a seventh communal snake hibernaculum, in accordance with an embodiment; and
(15) FIG. **8** is a flow chart of a method of installing a communal snake hibernaculum.
DETAILED DESCRIPTION
(16) Various apparatus or processes will be described below to provide an example of each claimed embodiment. No example described below limits any claimed embodiment and any claimed embodiment may cover processes or apparatuses that differ from those described below. The claimed embodiments are not limited to apparatuses or processes having all of the features of any one apparatus or process described below or to features common to multiple or all of the apparatus or processes described below.
(17) Referring now to FIG. **1**, illustrated therein is an example of a communal snake hibernaculum **100**. The communal snake hibernaculum **100** includes a communal chamber **102**. The communal chamber **102** provides a subterranean space for a plurality of snakes to overwinter together. The hibernaculum **100** provides a life zone **102***a* within the communal chamber **102**. The life zone **102***a* is configured to allow snakes to survive winter in a subterranean space that remain aerobic and does not freeze or flood completely. It will be appreciated that snakes can dive short lengths of time (e.g., a few hours) underwater, depending on temperature, by holding their breath, the length of time depending on the species and/or size of the snake. Therefore, a life zone can briefly flood for a short period of time (e.g., due to flash flooding caused by heavy rain). However, a life zone must not freeze because snakes cannot survive winter in a frozen state. Survival following brief exposures to freezing air temperatures outside a hibernacula has been documented for the red-sided garter snake, however this is not a continuous overwinter strategy. Mass die offs of Red-sided garter snakes have been documented from flooding and freezing of their hibernacula. In some examples, the life zone is above a groundwater level at the hibernaculum site. In some examples, the life zone is a zone between a frost line and a groundwater level at the hibernaculum site.
(18) The chamber **102** also includes a water zone **102***b* configured to hold water through the winter

to allow snakes to rehydrate. Snakes that choose a hibernaculum that is too dry are at risk of dehydration, and prolonged exposure to dry conditions can cause death. Some snakes can maintain water balance and avoid water loss due to thick skin and remaining in a coiled and still position to reduce water loss through evaporation. However, the presence of water within a hibernaculum provides an opportunity for the snake to re-hydrate periodically and maintain water balance.
(19) A water outlet in a sidewall of the chamber **102** helps to control water levels within the chamber **102**. When water rises in their hibernacula, e.g., from storm events, snakes may choose to dive and hold their breath for extended periods or climb higher up in the hibernacula towards the ground surface thus risking exposure to freezing temperatures to avoid submersion or contact with the water. Snakes that remain within the frost layer will die from freezing and snakes that remain underwater too long will drown. The water outlet allows water entering the chamber from above-ground water sources (e.g., storm events or melting) to freely pass through the sidewall to exit the chamber. In some examples, the water outlet is a port through the sidewall. In some examples, the water outlet includes a passage to carry water away from the hibernaculum. In some examples, the water outlet is above a groundwater level at the hibernaculum site.
(20) In some examples, the communal chamber **102** is an open system in that the water therein is in direct contact with the groundwater (e.g., through water permeable walls) to provide thermal stability within the chamber **102**. In some examples, the communal chamber **102** is a closed system in that the water contained therein is isolated from direct contact with groundwater (e.g., through water tight walls) and thermal stability is maintained using, e.g., insulation and air-movement limitation.
(21) The communal chamber **102** has a lower end **104** and an upper end **106**. Referring to FIG. **1**, in the exemplary chamber **102** a vertical axis **108** extends between the lower end **104** and the upper end **106**. The lower end **104** is below a ground level **110** (e.g., an average ground level around the hibernaculum apart from any overburden built over the hibernaculum). A lower end below a ground level provides a subterranean space. In some examples, the lower end **104** is below a frost level at the hibernaculum site, although in some examples temperature is maintained above freezing in other ways. While the entire chamber **102** may be below the ground level, in some examples the upper end is above the ground level. An upper end above the ground level provides easier access to the chamber (e.g., for snakes or for researchers interacting with the hibernaculum), reduces the chances of flooding, allows for easier air movement and ventilation, and/or allows for easier installation of the hibernaculum. Referring to FIG. **1**, the exemplary upper end **106** is above the ground level **110**. The upper end **106**, when above the ground level **110**, is insulated. Insulating the upper end **106** reduces heat transfer between the communal chamber and the ambient environment. Any suitable insulation may be used. Referring to FIG. **1**, the exemplary upper end **106** is insulated by an overburden **112**. The overburden **112** may include soil and/or rocks, such as rock ledges **194**.
(22) The chamber **102** may be any suitable shape, such as generally cylindrical, generally cuboid, or irregular in shape. Referring to FIG. **1**, the exemplary chamber **102** is generally cylindrical. Referring to FIG. **1**, the exemplary chamber **102** has a lower wall **114** at the bottom end and an upper wall **116** at the upper end with a sidewall **118** extending between the lower wall **114** and the upper wall **116**. The sidewall **118** may be any suitable shape, such as generally circular in cross section for a generally cylindrical chamber or generally rectangular in cross section for a generally cuboid chamber. It will be appreciated that the lower wall **114** and/or upper wall **116** may be, e.g., generally planar, concave, or irregular in shape.
(23) The exemplary sidewall **118** is a wall of a container **120** that is installed in the ground at the hibernaculum site **122**. The container **120** provides stability to the shape of the chamber **102** to prevent chamber collapse or erosion. In some examples, the container **120** is pre-formed before installation in the ground (e.g., a plastic drum transported to the site and installed), though in some examples the container may be formed at the hibernaculum site **122** (e.g., concrete pored at the site). In some examples, one or both of the upper wall **116** and the lower wall **114** are also part of

the container **120**.

(24) The walls forming the lower end **104** of the chamber are resistant to the passage of water. In use, the walls forming the lower end **104** of the chamber **102** form a barrier between water within the chamber **102** and groundwater around the chamber **102**. The walls forming the lower end **104** may be formed of any suitable material, such as concrete, cement, or plastic. In some examples, the walls forming the lower end **104** prevent the free movement of water but are not water-tight (e.g., walls formed of a material that is not water-proof and/or a sidewall **116** that is not sealed to a lower wall **114**). However, in some examples, the walls forming the lower end **104** are water-tight. Referring to FIG. **1**, the walls forming the lower end **104** are walls of a cement tank with a flat bottom wall and cylindrical sidewall cast together as a unitary body.

(25) The communal chamber **102** includes a snake entrance **130**. The snake entrance **130** opens from an exterior of the hibernaculum **100** into the chamber **102**. Referring to FIG. **1**, the exemplary snake entrance **130** includes a passage **132** extending between an intake end **134** opening to an exterior of the hibernaculum **100** and an outlet end **136** opening to an interior of the communal chamber **102**. It will be understood that, in some examples, a snake entrance **130** can also be used as a snake exit. In some examples, more than one snake entrance **130** is included. In some examples, at least three snake entrances **130** are included. Where a plurality of snake entrances are included, the snake entrances **130** may all be generally the same or a hibernaculum **100** may include two or more different types of entrances. Different types of snake entrances **130** may differ based on entry method (e.g., passive vs. manual) and/or differ in the snake type that they are configured to be used by (e.g., passage transverse size, passage length, passage entrance shape, passage entrance location, and/or passage wall material).

(26) The communal chamber **102** also includes a plurality of snake support shelves **140** within the chamber **102**. The snake support shelves **140** are arranged within the chamber **102** at a plurality of vertically-spaced locations between the upper end **106** and the lower end **104**. A plurality of vertically-spaced support shelves allows a snake to move up and down within the chamber **102** as desired to find a location suitable for the snake. A snake may move down within the chamber **102** to find a warmer location within the chamber **102**, e.g., during a cold mid-winter period. A snake may move down within the chamber **102** to access the water within the chamber **102** to rehydrate. A snake may move up within the chamber **102** to avoid drowning when the water level within the chamber rises. The snake shelves **140** form resting surfaces on which the snakes may rest, and the resting surface may be horizontal and/or angled. Referring to FIG. **1**, the exemplary snake support shelves **140** form resting surfaces which are angled relative to the vertical axis to facilitate snake movement from one resting surface to another. The snake support shelves **140** may be formed from any suitable material arranged in the chamber **102** (e.g., dropped into the container **120**), such as large rocks, pieces of wood, concrete blocks, and/or cinder blocks.

(27) The communal snake hibernaculum **102** includes a water outlet **150**. The water outlet **150** is provided to prevent the communal chamber **102** from flooding. Water may enter the hibernaculum **100** through, e.g., a water inlet (e.g., an inlet passage for seasonally freshening the water, as described elsewhere herein), filtering through the overburden and into the chamber at the upper end **106** if the upper end **106** is not water-tight, and/or seeping through the walls of the lower end **104** if the walls are not water-tight. According, an increase in water level within the chamber **102** may be driven by increases in the level of groundwater and/or by above-ground water sources (e.g., rain or melting snow). In some examples, the water outlet **150** is provided to provide drainage when water-level increases are driven by above-ground water sources.

(28) In some examples, the water outlet **150** may be a simple port in a wall of the chamber **102**, however in some examples the water outlet **150** includes a passage to carry water away from the chamber **102**. Referring to FIG. **1**, the exemplary water outlet **150** includes a passage **152** extending between an intake end **154** and a discharge end **156**. The intake end **154** is in the sidewall **118** of the communal chamber **102**. The intake end **154** opens into the chamber **102** at a location

that is above the lower end **104** and below at least one of the snake support shelves **140** to prevent flooding of the communal chamber **102**. In use, water is held in the lower end of the communal chamber **102** for snake rehydration, but the water level is prevented from rising over all of the support shelves **140** by the water outlet **150**. In some examples, the intake and/or discharge ends of the passage **152** are screened to prevent snakes from entering the passage **152**.
(29) Any suitable water outlet **150** may be used. In some examples, the water outlet may be an active (e.g., powered) water outlet, such as a water outlet that includes a pump to remove water as needed to maintain a water level below at least one of the snake support shelves. However, in some examples, the water outlet **150** is a passive water outlet. A passive water outlet **150** uses gravitational force to drain water from the chamber **102**. In some examples, the discharge end **156** is below the intake end **154**. A passive water outlet **150** with a passage **152** has a discharge end **156** shaped and/or positioned to discharge water from the water outlet **150**. In some examples, the discharge end **156** is above ground (e.g., into an adjacent depression in the ground, as described further below). In some examples, the discharge end **156** is underground (e.g., into a drainage bed, such as a bed of gravel). A water outlet **150** with an underground discharge end **156** has an intake end **154** above a groundwater level **160**, and optionally the discharge end **156** is also above the groundwater level **160**.
(30) Referring to FIG. **1**, a groundwater level **160** at the exemplary hibernaculum site **122** fluctuates between a seasonal high level **162** and a seasonal low level **164**. The intake end **154** of the exemplary water outlet **150** of FIG. **1** is above the seasonal low level **164**. In some examples, the intake end **154** is also above the seasonal high level **162**. The exemplary discharge end **156** of FIG. **1** is also above the seasonal low level **164**. In some examples, the discharge end **156** is also above the seasonal high level **162**. In some examples, the intake end **154** is between 30 cm and 70 cm below the ground level **110**, between 40 cm and 60 cm below the ground level **110**, or about 50 cm below the ground level **110**. In some examples, the discharge end **156** is between 30 cm and 70 cm below the ground level **110**, between 40 cm and 60 cm below the ground level **110**, or about 50 cm below the ground level **110**. The discharge end **156** of the exemplary water outlet **150** of FIG. **1** is at a depth **166** of about 50 cm below the ground level **110**. The discharge end **156** of the water outlet **150** is below the intake end **154** to encourage water flow through the outlet **150**.
(31) In some examples, the hibernaculum includes a monitoring well **170**. The monitoring well **170** extends into the chamber **102** from an exterior of the hibernaculum **100** to permit users (e.g., researchers) to access the interior of the chamber **102** to, e.g., insert or access sensors. Referring to FIG. **1**, the exemplary hibernaculum **100** includes a monitoring well **170** extending into the chamber **102** through the upper end **106**. In some examples, the well **170** includes a generally vertically extending casing **171** with apertures **176** spaced along its length. In some examples, the casing is generally linear. A linear casing allows a floating water level sensor to move freely along the length of the well. In some examples, a well **170** is supported by a rebar support **173**, which may extend along the length of the well **170** and be anchored at the top and bottom ends. Referring to FIG. **1**, the exemplary well **170** is a linear, vertical pipe containing a floating sensor.
(32) In some examples, the hibernaculum **100** includes a monitoring system **172**. The monitoring system **172** includes a sensor **174** operable to sense a water level, temperature, and/or water quality. in some examples, the sensor **174** is attached to a cable **175** located within the monitoring well **170**. The cable **175** may be used to lower and raise the sensor **174**, e.g., for user access. In some examples, the sensor **174** is suspended within the well by the cable **175**. In some examples, the sensor **174** is a logger that stores detected data locally to be accessed by a user at the hibernaculum site **122** (e.g., by plugging into the logger). In some examples, the sensor **174** includes a transmitter to transfer data to a remote location (e.g., includes an antenna). Referring to FIG. **1**A, the exemplary monitoring system **172** includes a water level logger sensor **174***a* in the well **170**, a temperature logger **174***b*, and an air barometer **174***c*. The exemplary water level logger sensor **174** of FIG. **1**A also logs temperature data. In some examples, the sensor **174** is contained in the well

170. The exemplary sensor **174** is contained in the well **170**, and the exemplary well **170** extends down into the chamber **102** and is in thermal and fluid communication with the interior of the chamber **102** via a plurality of inlets **176** along the length thereof. The inlets **176** are small inlets to prevent snake access, such as less than 10 mm in diameter, less than 6 mm in diameter, or less than 5 mm in diameter. The well **170** is installed with at least some inlets **176** below a frost line **183**. Referring to FIGS. **1**A, the exemplary well **170** has an openable access end **178** selectively closed by a cap **180**. The cap **180** closes off the openable access end **178** to prevent animals from entering the casing **171**, while at the same time leaving the well **170** selectively openable for access by users (e.g., researchers). In some examples, the bottom end of the casing **171** opposite the access end **178** is also closed and/or screened to prevent animal access. The exemplary monitoring system **172** of FIG. **1**A also includes a lock **181** to prevent unauthorized access to the sensors **174**.
(33) In some examples, the monitoring system **172** includes a plurality of sensors **174**, such as a floating sensor to provide water level data (and optionally temperature data at the water surface) and one or more fixed position sensors to provide temperature data for a fixed location within the chamber **102**. In some examples, the floating sensor and the one or more fixed sensors are all contained in the well **170** for ready access. Referring to FIG. **1**, in some examples a monitoring system **172** includes a floating sensor in the well **170**, and one or more of a fixed sensor in a first location **182***a* at a bottom end of the well **170** to provide temperature data for the bottom end of the chamber **102**, a fixed sensor in the well **170** in a second location **182***b* at the upper end of the chamber **102** to provide temperature data for the top end of the chamber **102**, and a fixed sensor in a third location **182***c* at a top end of the well **170** to provide temperature data for the ambient environment outside the hibernaculum.
(34) Referring now to FIGS. **2**A and **2**B, the exemplary communal snake hibernaculum **200** is similar in many respects to communal snake hibernaculum **100**, and like components are indicated by like reference numbers incremented by 100.
(35) The exemplary communal snake hibernaculum **200** of FIGS. **2**A and **2**B includes a communal chamber **202** with a lower end **204** below a ground level **210** and an upper end **206** above the ground level **210** and insulated by an overburden **212**. The chamber **202** includes a sidewall **218**, an upper wall **216**, and a lower wall **214**. The exemplary communal chamber **202** of FIGS. **2**A and **2**B includes a plurality of snake entrances **230**, a plurality of snake support shelves **240**, a monitoring system **272**, and a water outlet **250**.
(36) As noted above, in some examples, a water outlet has an above-ground discharge end. Referring to FIGS. **2**A and **2**B, the exemplary water outlet **250** has an above-ground discharge end **256**. The exemplary discharge end **256** of FIGS. **2**A and **2**B opens into a swale **284**. The exemplary swale **284** of FIGS. **2**A and **2**B is adjacent the hibernaculum **200** to reduce the necessary length of the passage **282**, although in some examples the passage **282** may be a long passage to, e.g., carry water to a distant depression in the ground. In some examples, the dirt removed to create the swale is used to form the overburden over the upper end of the chamber **202**.
(37) In some examples, a snake hibernaculum **200** includes different classes of snake entrances. Different classes of snake entrances may be configurated for different types of snakes (e.g., different species and/or different sizes) and/or different entry methods (e.g., of the snake's own volition or manually introduced by a user).
(38) Referring to FIGS. **2**A and **2**B, the exemplary hibernaculum **200** includes a first snake entrance **230***a* for a first type of snake and a second snake entrance **230***b* for a second type of snake, the second type of snake being different from the first type of snake. In some examples, the second type of snake is a different species than the first type of snake (e.g., massasauga rattlesnake vs. fox snake). In some examples, the second type of snake is a different diameter than the first type of snake. In some examples, entrances for different types of snake differ from one another by minimum transverse passage area (e.g., different minimum diameters for cylindrical passages), wall construction (e.g., amount of traction), and/or location (e.g., intake end height above ground

level or intake end size and shape). Referring to FIGS. **2**A and **2**B, the exemplary first snake entrance **230***a* has a larger diameter than the exemplary second snake entrance **230***b*. Different classes of entrance may also or alternatively differ based on intake end characteristics, such as whether the intake end opening to the outside of the hibernaculum is overhung by rock ledges **294** or other overburden material. Referring to FIGS. **2**A and **2**B, the exemplary snake entrances **230** extend beyond the overburden material.

(39) Referring to FIGS. **3**A to **3**C, the exemplary communal snake hibernaculum **300** is similar in many respects to communal snake hibernaculum **100**, and like components are indicated by like reference numbers incremented by 200.

(40) The exemplary communal snake hibernaculum **300** of FIGS. **3**A to **3**C includes a communal chamber **302** with a lower end **304** below a ground level **310** and an upper end **306** above the ground level **310** and insulated by an overburden **312**. The chamber **302** includes a sidewall **318**, an upper wall **316**, and a lower wall **314**. The exemplary communal chamber **302** of FIGS. **3**A to **3**C includes a plurality of snake entrances **330**, a plurality of snake support shelves **340**, a monitoring system **372**, and a water outlet **350** to a swale **384**.

(41) Referring to FIGS. **3**A to **3**C, the exemplary snake support shelves **340** are formed by pipe angular portions, such as half-pipes. The exemplary angular positions of FIGS. **3**A to **3**C are portions of plastic pipes, such as polyvinyl chloride (PVC) pipes.

(42) As mentioned elsewhere herein, in some examples the sidewall **318** and the one or both of the upper wall **316** and the lower wall **314** that are also part of the container **320** are secured to one another in a water-tight connection (e.g., the container **320** may be formed of a unitary body comprising the sidewall **318** and one or both of the upper wall **316** and the lower wall **314**). For example, the container **320** may include a plastic (e.g., polyethylene) cylindrical sidewall from a water storage tank (e.g., from a Bushman™ water storage tank). Referring to FIGS. **3**A to **3**C, the exemplary hibernaculum **300** includes a unitary container **320** forming the lower wall **314** and the sidewall **318**. In the exemplary hibernaculum **300** of FIGS. **3**A to **3**C, the container **320** is a cylindrical plastic tank, and the upper wall **316** is formed by a removable lid over an open end of the plastic tank. The exemplary plastic drum of FIGS. **3**A to **3**C is a molded polyethylene tank.

(43) In some examples in which the container **320** is formed of a material that is lighter than concrete or cement, such as plastic, the hibernaculum further includes additional insulation and/or stability support.

(44) In some examples, the sidewall **318** is insulated. Referring to FIGS. **3**A to **3**C, the exemplary sidewall **318** is insulated by a layer of insulation **390**. The insulation **390** may be interior or exterior to the container sidewall. The exemplary insulation **390** of FIGS. **3**A to **3**C is exterior insulation against the sidewall **318**. The insulation can be any suitable material. In some examples, the insulation is formed of a natural fiber. In some examples, the insulation **390** is formed of hemp. While the insulation **390** extends over the bottom wall **314** and/or top wall **316** in some examples, in some examples the top wall **316** is only insulated by the overburden and the bottom wall **314** is uninsulated.

(45) In some examples, the hibernaculum **300** includes a support layer under the container **320**. Referring to FIGS. **3**A to **3**C, the exemplary container **320** is supported on a support layer **392**. The support layer **392** forms a hard surface to support the container **320**, e.g., against rocking and/or settling. The exemplary support layer **392** of FIGS. **3**A to **3**C is a masonry support layer. In some examples, a masonry support layer is formed of rocks, bricks, and/or tiles. The exemplary support layer **392** of FIGS. **3**A to **3**C is formed of a layer of laterally-abutting patio stones.

(46) In some examples in which the chamber **304** is a closed system relative to groundwater (e.g., the bottom end **304** is water-tight) the water contained in the chamber **302** is freshened by water coming in through an upper end of the chamber **304** and/or through a water inlet passage opening into the chamber. In some examples, water filters into the upper end of the chamber through a porous upper end, such as by filtering through the overburden and through a water-permeable cover

on which the overburden rest. Referring to FIGS. **3**A to **3**C, the exemplary hibernaculum **300** includes a water-permeable upper wall **316** to close the open upper end of the plastic drum that forms the sidewall **318** and bottom wall **314** of the chamber **302**. The water-permeable upper wall **316** allows a water flow **388** to filter through the overburden and into the upper end **306** of the chamber **302**. In some examples, the water-permeable upper wall **316** is a removeable cover.

(47) In some examples, a communal snake hibernaculum **300** includes different types of entrances for different entry methods. Referring to FIGS. **3**A to **3**C, the exemplary hibernaculum **300** includes a third snake entrance **330***c* which is a passive snake entrance and a fourth snake entrance **330***d* which is a manual snake entrance. A passive snake entrance is provided to allow a snake to enter the hibernaculum of its own volition. A manual entrance is provided to allow a user (e.g., a researcher) to insert a snake into the hibernaculum. In use, users may introduce snakes into the hibernaculum to familiarize the snakes with the hibernaculum and promote use of the hibernaculum. As snakes become familiar with the hibernaculum, they may be more likely to use the hibernaculum of their own volition.

(48) In some examples, a passive entrance is always open while a manual entrance is selectively closed by a door (e.g., a lid covering the intake end). In some examples, a passive entrance is shaped to inhibit rain entry (e.g., intake port directed laterally or downward, intake end and outlet end at generally equal elevations, and/or intake end raised above the ground level **310** to inhibit over-land water entrance). In some examples, a manual entrance is shaped to cause a snake introduced into the intake end to exit into the chamber (e.g., low-traction walls, intake end at an elevation that is above the outlet end, and/or walls forming a generally continuous ramp). In some examples, the manual entrance **330***d* has an intake end outside the chamber at an elevation above the discharge end inside the chamber. In some examples, the manual entrance **330***d* includes a passage that is extends downward continuously along the entire length of the passage to encourage snakes introduced into the passage to continue down the passage to the end. In some examples, the intake end of the manual passage **330***d* is generally vertical and/or smooth sided to prevent snakes that have been introduced into the manual entrance **330***d* from exiting through the manual entrance **330***d*. The exemplary manual entrance **330***d* of FIGS. **3**A to **3**C includes a vertically-extending linear passage **332**, e.g., to drop a snake into the communal chamber **302**. In some cases, a manual snake entrance may be sealed up or converted to a passive snake entrance after a period of time (e.g., a year or a few years), e.g., after snakes have become used to using the hibernaculum. The manual entrance may be converted by, e.g., removing a vertical and/or smooth-sided portion of a passage.

(49) Referring to FIGS. **4**A and **4**B, the exemplary communal snake hibernaculum **400** is similar in many respects to communal snake hibernaculum **100**, and like components are indicated by like reference numbers incremented by 300.

(50) The exemplary communal snake hibernaculum **400** of FIGS. **4**A and **4**B includes a communal chamber **402** with a lower end **404** below a ground level **410** and an upper end **406** above the ground level **410** and insulated by an overburden **412**. The chamber **402** includes a sidewall **418**, an upper wall **416**, and a lower wall **414**. The exemplary communal chamber **402** of FIGS. **4**A and **4**B includes a plurality of snake entrances **430**, a plurality of snake support shelves **440**, a monitoring system **472**, and a water outlet **450**.

(51) The exemplary hibernaculum **400** of FIGS. **4**A and **4**B includes a framework **496** to support the upper wall **416**. The hibernaculum **400** maintains a chamber **402** against collapse. The framework **496** is provided to hold up the upper wall **416** and the weight of the insulating overburden, e.g., where the container is formed of a material (e.g., plastic) that is not strong enough to support the weight of the overburden **412**. Any suitable framework may be used, and the framework may be anchored in or against any suitable support layer. Referring to FIGS. **4**A and **4**B, the exemplary framework **496** includes generally vertical support beams **496***a* extending generally parallel to the sidewall **418** and anchored in the earth below the support layer **492** (e.g., a

layer of patio stones in an insulation layer **490**). In some examples, support beams are braced against and/or anchored in the support layer **492**. In some examples, the support beams are wooden beams.

(52) In some examples, the lowest shelf **440** is arranged against the bottom wall **414** to prevent snakes from moving between the lowest shelf and the bottom wall and becoming trapped. In some examples, the lowest shelf **440** is anchored to the bottom wall **414**. Referring to FIGS. **4**A and **4**B, the exemplary chamber **402** includes a cedar wood insert **498** within the bottom end **404**. The cedar wood insert **498** is arranged to be in contact with water that is within the chamber **402**. In use, the cedar wood lowers the pH of the water in the chamber. Referring to FIGS. **4**A and **4**B, the exemplary cedar wood insert **498** forms the lowest shelf **440** within the chamber **402**.

(53) Referring to FIGS. **5**A and **5**B, the exemplary communal snake hibernaculum **500** is similar in many respects to communal snake hibernaculum **100**, and like components are indicated by like reference numbers incremented by 400.

(54) The exemplary communal snake hibernaculum **500** of FIGS. **5**A and **5**B includes a communal chamber **502** with a lower end **504** below a ground level **510** and an upper end **506** above the ground level **510** and insulated by an overburden **512**. The communal chamber **502** includes a sidewall **518**, an upper wall **516**, and a lower wall **514**. The exemplary communal chamber **502** of FIGS. **5**A and **5**B includes a plurality of snake entrances (including a manual entrance **530***d* and a passive entrance **530***c*), a plurality of snake support shelves **540**, a monitoring system **572**, and a water outlet **550**. The exemplary hibernaculum also includes a support framework **596** with support beams **596***a*.

(55) The exemplary overburden **512** of FIGS. **5**A and **5**B includes a water collection feature **558**. The exemplary water collection feature **558** is a depression in the overburden **512** to collect water, e.g., melt water or rain water. The exemplary collection feature **558** of FIGS. **5**A and **5**B overlies the upper end **506** of the communal chamber **502** to encourage water to flow **588** through the overburden and into the chamber **502** to freshen the water within the chamber **502**.

(56) The exemplary snake support shelves **540** of FIGS. **5**A and **5**B are formed of masonry blocks **568**. Masonry blocks **568** may be, e.g., cinder blocks, concrete blocks, bricks, or blocks of natural stone. In some examples, a masonry blocks **568** has a cell **568***a* opening through the block. Cells opening through the blocks provide crannies for snake movement and occupation. The exemplary masonry blocks **568** of FIGS. **5**A and **5**B are cinder blocks, each with two open cells **568***a*. The exemplary snake shelves **540** of FIGS. **5**A and **5**B are formed by stacked cinder blocks.

(57) Referring to FIG. **6**, the exemplary communal snake hibernaculum **600** is similar in many respects to communal snake hibernaculum **100**, and like components are indicated by like reference numbers incremented by 500.

(58) The exemplary communal snake hibernaculum **600** of FIG. **6** includes a communal chamber **602** with a lower end **604** below a ground level **610** and an upper end **606** above the ground level **610** and insulated by an overburden **612**. The exemplary chamber **602** includes a sidewall **618**, an upper wall **616**, and a lower wall **614**. The exemplary communal chamber **602** of FIG. **6** includes a plurality of snake entrances (including a manual entrance **630***d* and a passive entrance **630***c*), a plurality of snake support shelves **640**, a monitoring system **672**, and a water outlet **650**.

(59) Referring to FIG. **6**, the exemplary snake hibernaculum **600** includes a water inlet **686**. The water inlet **686** is provided to allow water to enter the chamber **602** to freshen the water in the bottom end of the chamber. The water inlet **686** may be provided in addition to, or as an alternative, to other pathways of water entry such as seepage through the overburden and upper wall and/or seepage through the sidewall or between the sidewall and the bottom wall. The water inlet **686** may be provided to seasonally freshen the water in the chamber **602**.

(60) In some examples, the water inlet **686** may be a simple port in a wall of the chamber **102**, however in some examples the water inlet **686** includes a passage to carry water into the chamber **602**. Referring to FIG. **6**, the exemplary water inlet **686** includes a passage **686***a* extending between

an intake end **686***b* opening to an outside of the hibernaculum and a discharge end **686***c* opening into the communal chamber **602**.
(61) Any suitable water inlet **686** may be used. In some examples, the water inlet may be an active (e.g., powered) water outlet, such as a water inlet that includes a pump to add water as needed. However, in some examples, the water inlet **686** is a passive water inlet. A passive water inlet **686** uses gravitational force to drain water from a source into the chamber **602**. In some examples, the discharge end **686***c* is below the intake end **686***b*.
(62) The intake end **686***b* and the discharge end **686***c* may each be at any suitable location. The exemplary discharge end **686***c* of FIG. **6** opens through the sidewall **618** of the chamber **602**. In some examples, the discharge end **686***c* opens through another wall of the chamber **602**, such as the upper wall **616**. The exemplary discharge end **686***c* of FIG. **6** opens above the intake end **654** of the water outlet **650**. In some examples, the discharge end **686***c* of the water inlet is below the intake end **654** of the water outlet **650** (e.g., maintaining a water level in the chamber that is lower than the intake end of the water outlet or using hydrostatic pressure or pumping to raise the water level in the chamber **602** to the intake end **654** of the water outlet).
(63) In some examples, the intake end **686***b* of the water inlet **686** includes an above-ground intake port. In some examples, the intake end **686***b* opens to a seasonally available water supply to seasonally introduce new water to the communal chamber. Referring to FIG. **6**, the exemplary intake end **686***b* is above-ground and opens to an upper portion of a water-holding depression **626**. The water level **628** in the depression **626** fluctuates seasonally, and is below the intake end **686***b* during a first season (e.g., winter) and above the intake end **686***b* during a second season (e.g., spring and/or fall).
(64) Referring to FIG. **7**, the exemplary communal snake hibernaculum **700** is similar in many respects to communal snake hibernaculum **100**, and like components are indicated by like reference numbers incremented by **600**.
(65) The exemplary communal snake hibernaculum **700** of FIG. **7** includes a communal chamber **702** with a lower end **604** below a ground level **710** and an upper end **706** above the ground level **710**. The exemplary chamber **702** includes a sidewall **718**, an upper wall **716**, and a lower wall **714**. The exemplary communal chamber **702** of FIG. **7** includes a plurality of snake entrances **730** (including a manual entrance **730***d* and a passive entrance **730***c*) and a water outlet **750**. The exemplary outlet **750** is a 4 inch diameter port opening through the sidewall **718**.
(66) Referring to FIG. **7**, the exemplary chamber **702** is formed by a cuboid container **720**. The exemplary container **720** is a cement box having a screened manhole opening **724** in the top wall **716**. The screened manhole opening **724** allows water precipitation into the chamber **702**. In some examples, a cement container is sealed to be water-tight, e.g., by the application of a sealant layer or film.
(67) Referring to FIG. **8**, the exemplary method **800** of constructing a communal snake hibernaculum. The method **800** includes, at step **802**, installing a container (e.g., any of the containers disclosed herein) at a hibernaculum site. Step **802** includes installing the container to form a communal chamber with a subterranean water zone at a lower end of the chamber to hold rehydration water and an additional zone above the water zone and positioned to ordinarily remain aerobic, above a water level in the communal chamber, and above freezing such that the additional zone is a snake hibernaculum life zone.
(68) Step **802** further includes forming at least one snake entrance opening into the communal chamber from an exterior of the communal snake hibernaculum to provide snake access to the life zone. Step **802** may include installing the container such that an upper end of the container is above the ground level. Step **802** may include forming an overburden over the upper end of the container.
(69) At step **804**, the method **800** includes forming a water outlet opening through a sidewall of the container above the lower end of the chamber and below the life zone to prevent flooding of the life zone. In some examples, a level of groundwater outside the communal chamber fluctuates between

a seasonal high level during a first season and a seasonal low level during a second season, and the intake end of the water outlet is above the seasonal low level. In some examples, the intake end of the water outlet is also above the seasonal high level. In some examples, a discharge end of the water outlet is also above the seasonal low level and/or the seasonal high level.
(70) At step **806**, the method **800** includes forming a plurality of snake support shelves within the communal chamber at a plurality of vertically-spaced locations between the lower end and an upper end of the container, including at least one snake support shelf in the life zone. In some examples, step **806** includes arranging material within the chamber through an opening in an upper wall of the chamber, such as through the manhole opening **724** of the container **720** of the exemplary hibernaculum of FIG. **7**. In some examples, step **806** includes forming the plurality of snake support shelves includes arranging pieces of earthen material loosely within the communal chamber with snake passages therebetween. In some examples, the pieces of earthen material are cinder blocks each having at least one internal cell opening through the block.
(71) The present invention has been described here by way of example only. Various modification and variations may be made to these examples without departing from the scope of the invention, which is limited only by the appended claims.

Claims

1. A communal snake hibernaculum, comprising: a communal chamber installable at a hibernaculum site with a life zone within the chamber at a depth below a frost line and above a groundwater level, the communal chamber including a sidewall extending between a lower end and an opposing upper end with a water outlet opening through the sidewall between the lower end and the life zone to separate the life zone from a water zone at the lower end of the chamber and prevent flooding of the life zone when the communal chamber is installed at the hibernaculum site; a plurality of snake support shelves within the communal chamber at a plurality of positions spaced between the upper and lower ends of the chamber, including at least one snake support shelf within the life zone; and a snake entrance opening from an exterior of the communal snake hibernaculum into the communal chamber above the water outlet to provide snake access to the life zone; further comprising a monitoring well extending into the chamber from an exterior of the communal snake hibernaculum, the monitoring well generally vertically extending and configured to allow a sensor to be lowered and raised for user access to the sensor, the sensor operable to sense data including a water level, a temperature, and/or water quality, and log the data and/or transmit the data to a remote location.
2. The communal snake hibernaculum of claim 1, wherein the water outlet includes a passage extending between an intake end opening through the sidewall and a discharge end, the passage provided to carry water away from the hibernaculum when installed at the hibernaculum site.
3. The communal snake hibernaculum of claim 2, installed at the hibernaculum site with the water outlet passage having an above-ground discharge end.
4. The communal snake hibernaculum of claim 1, further comprising a water inlet opening into the communal chamber above the lower end of the communal chamber to supply water to fill the water zone.
5. The communal snake hibernaculum of claim 4, installed at the hibernaculum site with an intake end of the water inlet opening from a seasonally-available water supply to seasonally introduce new water to the communal chamber.
6. The communal snake hibernaculum of claim 1, installed at the hibernaculum site with the bottom end of the communal chamber extending below the ground level to a depth of at least **1** meter.
7. The communal snake hibernaculum of claim 1, installed at the hibernaculum site with the upper end above the ground level and insulated by an overburden.
8. The communal snake hibernaculum of claim 1, wherein at least a plurality of the snake support

shelves are formed by a plurality of blocks each having at least one internal cell opening through the block.
9. The communal snake hibernaculum of claim 1, wherein the communal chamber is sized to hold a plurality of snakes and has a volume of at least **7,500** cubic centimeters.
10. The communal snake hibernaculum of claim 1, wherein the communal snake hibernaculum is a multi-species hibernaculum and has a plurality of snake entrances including a first snake entrance sized for a first snake species and a second snake entrance sized for a second snake species, the first snake entrance being a different size than the second snake entrance.
11. A communal snake hibernaculum, comprising: a communal chamber having a lower end below a ground level at a hibernaculum site with a water zone at the lower end to hold rehydration water and a snake hibernaculum life zone above the water zone, the communal chamber having a snake entrance providing snake access to the life zone and including snake support shelves within the communal chamber at a plurality of vertically-spaced locations, the snake support shelves including at least one shelf in the life zone; and a water outlet opening through a sidewall of the communal chamber above the lower end of the communal chamber and below the life zone to prevent flooding of the life zone, and wherein the sidewall is a wall of a container installed at the hibernaculum site; further comprising a monitoring well extending into the chamber from an exterior of the communal snake hibernaculum, the monitoring well generally vertically extending and configured to allow a sensor to be lowered and raised for user access to the sensor, the sensor operable to sense data including a water level, a temperature, and/or water quality, and log the data and/or transmit the data to a remote location.
12. The communal snake hibernaculum of claim 11, wherein the container further comprises a bottom wall, and the sidewall and bottom wall form the lower end below the ground level.
13. The communal snake hibernaculum of claim 12, wherein the container is a plastic drum and the sidewall and bottom wall are part of a unitary body.
14. The communal snake hibernaculum of claim 11, wherein the life zone is above a groundwater level at the hibernaculum site.
15. The communal snake hibernaculum of claim 14, wherein the snake hibernaculum life zone is at a depth that is below a frost line at the hibernaculum site.
16. A method of constructing a communal snake hibernaculum, comprising: installing a container at a hibernaculum site to form a communal chamber with a subterranean water zone at a lower end of the chamber to hold rehydration water and an additional zone above the water zone and positioned to ordinarily remain aerobic, above a water level in the communal chamber, and above freezing such that the additional zone is a snake hibernaculum life zone; forming at least one snake entrance opening into the communal chamber from an exterior of the communal snake hibernaculum to provide snake access to the life zone; forming a water outlet opening through a sidewall of the container above the lower end and below the life zone to prevent flooding of the life zone; and forming a plurality of snake support shelves within the communal chamber at a plurality of vertically-spaced locations between the lower end and an upper end of the container, including at least one snake support shelf in the life zone; and forming a monitoring well extending into the chamber from an exterior of the communal snake hibernaculum, the monitoring well generally vertically extending and configured to allow a sensor to be lowered and raised for user access to the sensor, the sensor operable to sense data including a water level, a temperature, and/or water quality, and log the data and/or transmit the data to a remote location.
17. The method of claim 16, wherein installing the container at the hibernaculum site includes installing the container with an upper end of the container above the ground level, the method further comprising forming an overburden over the upper end of the container.
18. The method of claim 16, wherein a level of groundwater outside the communal chamber fluctuates between a seasonal high level during a first season and a seasonal low level during a second season, and the water outlet opens into the communal chamber at a position that is above

the seasonal low level.

19. The method of claim 16, wherein forming the plurality of snake support shelves includes arranging cinder blocks within the communal chamber, each cinder block having at least one internal cell opening through the block.